Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 2

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.503200
<D> / site : 0.250533

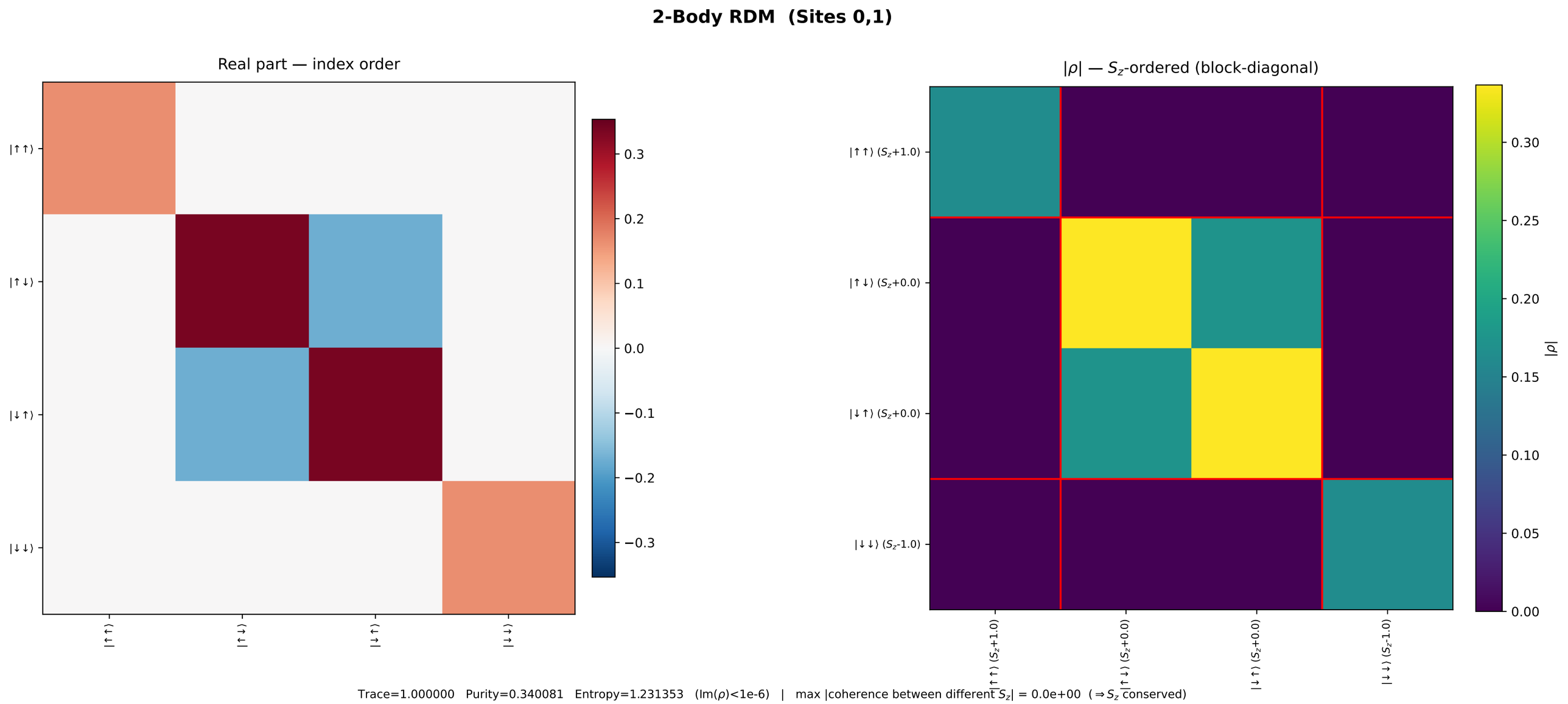
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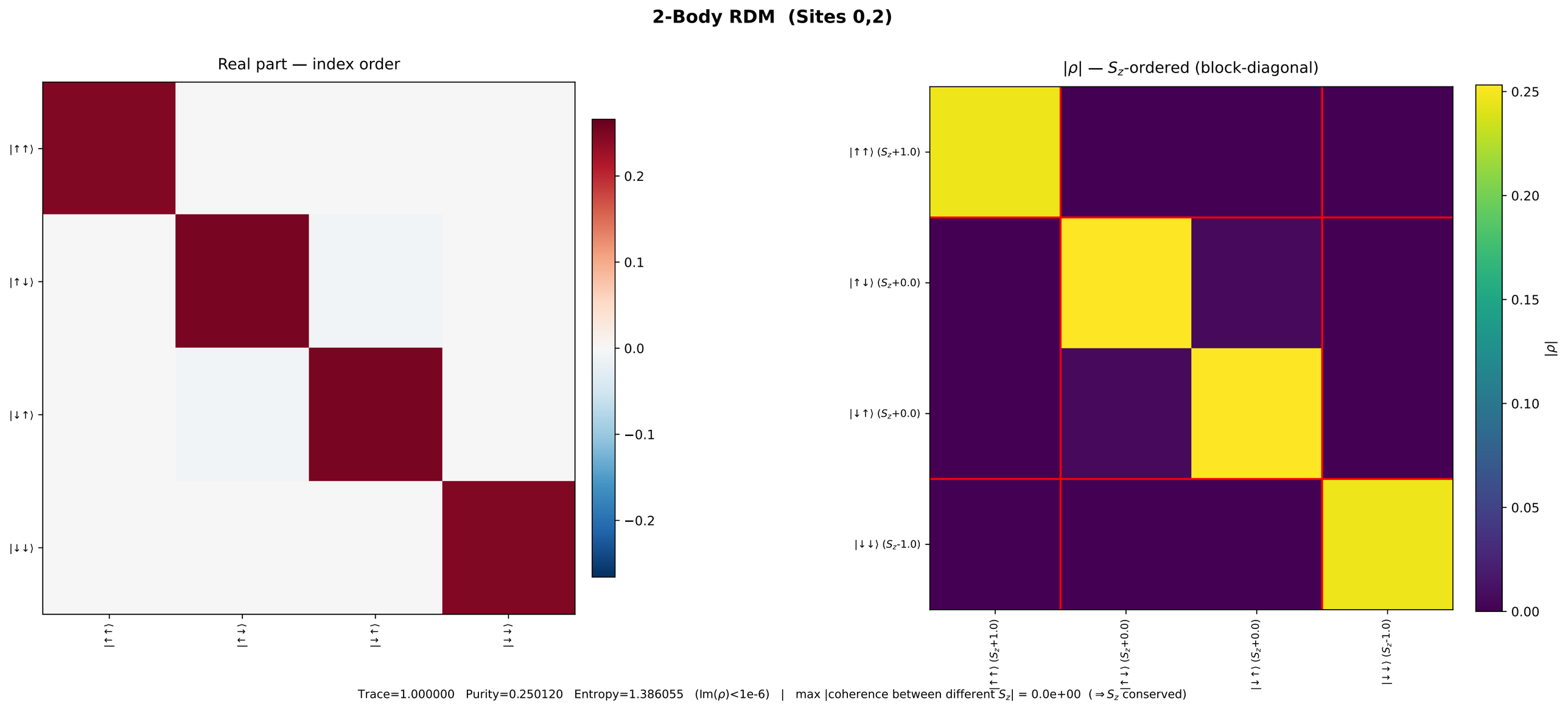
Body order : 2
Clusters   : 15

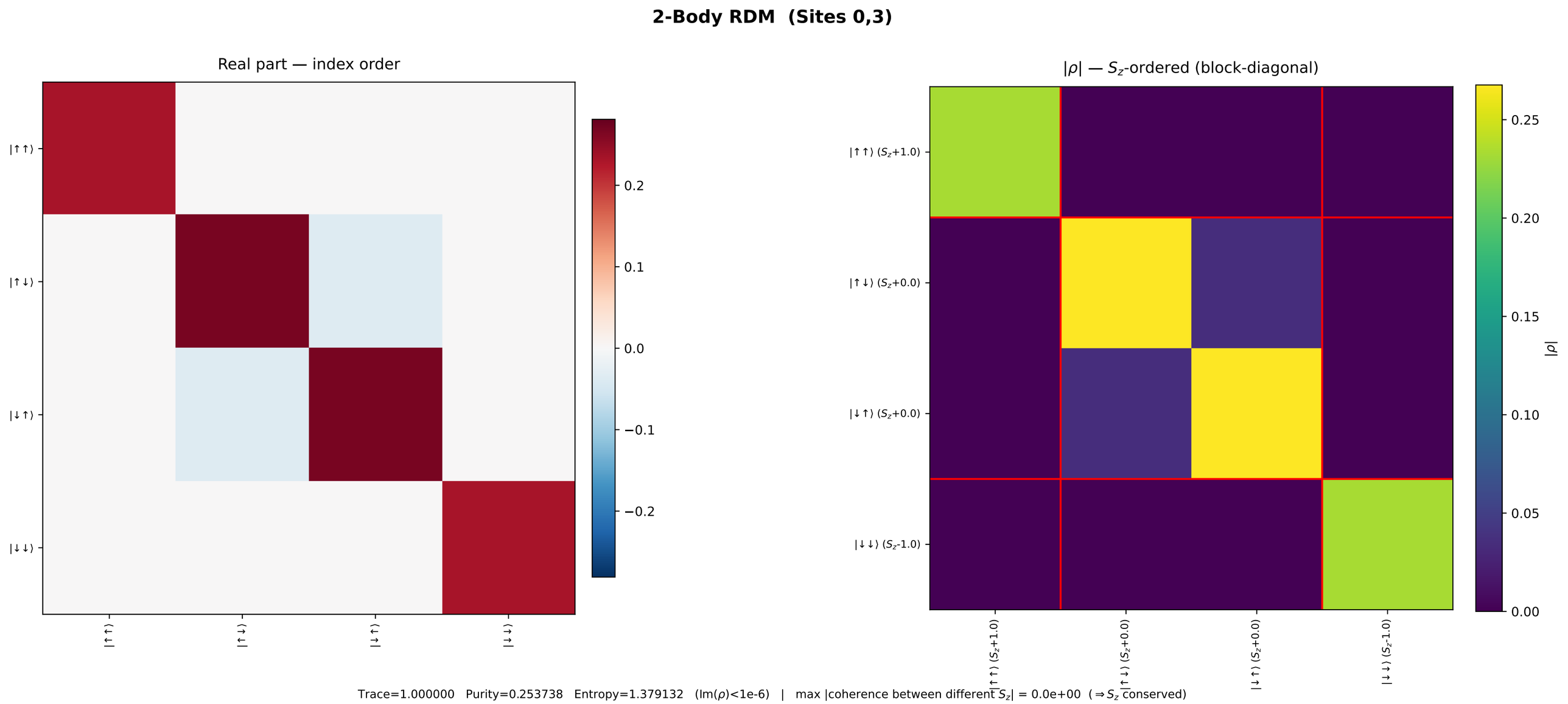
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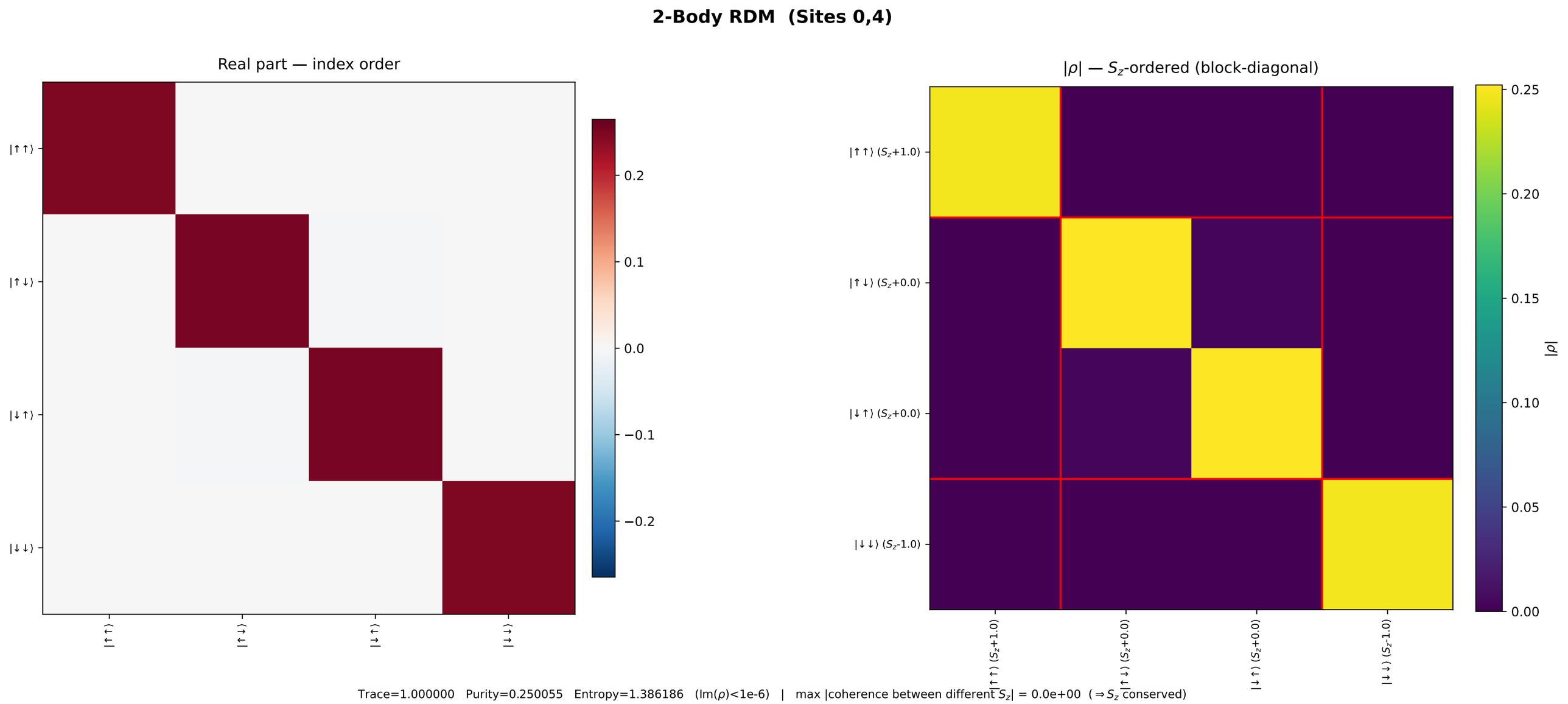
Summary — 2-Body RDMs (Benzene, CAS(6,6))

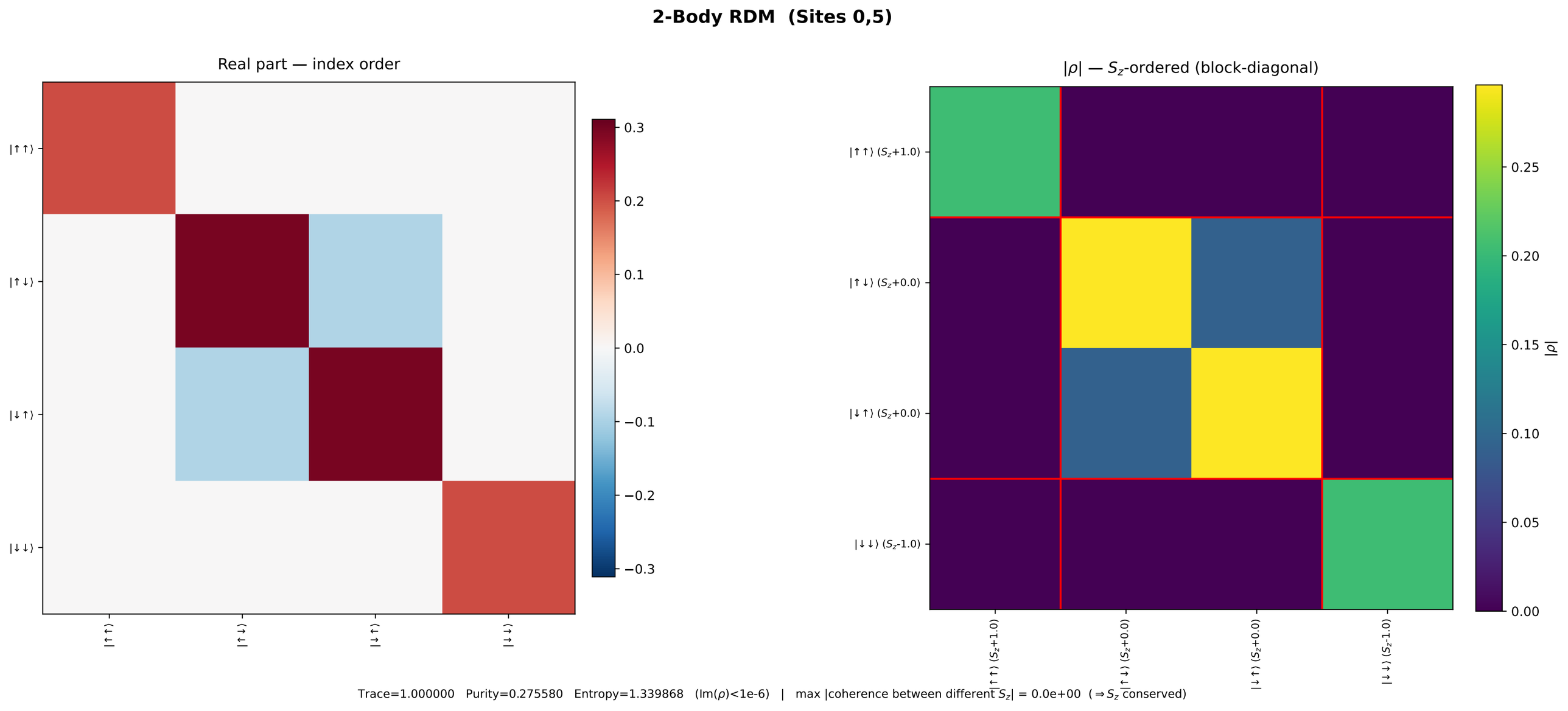
Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1	0.340081	1.231353	—	—	—	—	MIXED
0,2	0.250120	1.386055	—	—	—	—	MIXED
0,3	0.253738	1.379132	—	—	—	—	MIXED
0,4	0.250055	1.386186	—	—	—	—	MIXED
0,5	0.275580	1.339868	—	—	—	—	MIXED
1,2	0.253418	1.379733	—	—	—	—	MIXED
1,3	0.251937	1.382541	—	—	—	—	MIXED
1,4	0.277197	1.337048	—	—	—	—	MIXED
1,5	0.250792	1.384743	—	—	—	—	MIXED
2,3	0.272285	1.345641	—	—	—	—	MIXED
2,4	0.250087	1.386122	—	—	—	—	MIXED
2,5	0.330756	1.246704	—	—	—	—	MIXED
3,4	0.354668	1.207474	—	—	—	—	MIXED
3,5	0.252185	1.382068	—	—	—	—	MIXED
4,5	0.254680	1.377367	—	—	—	—	MIXED

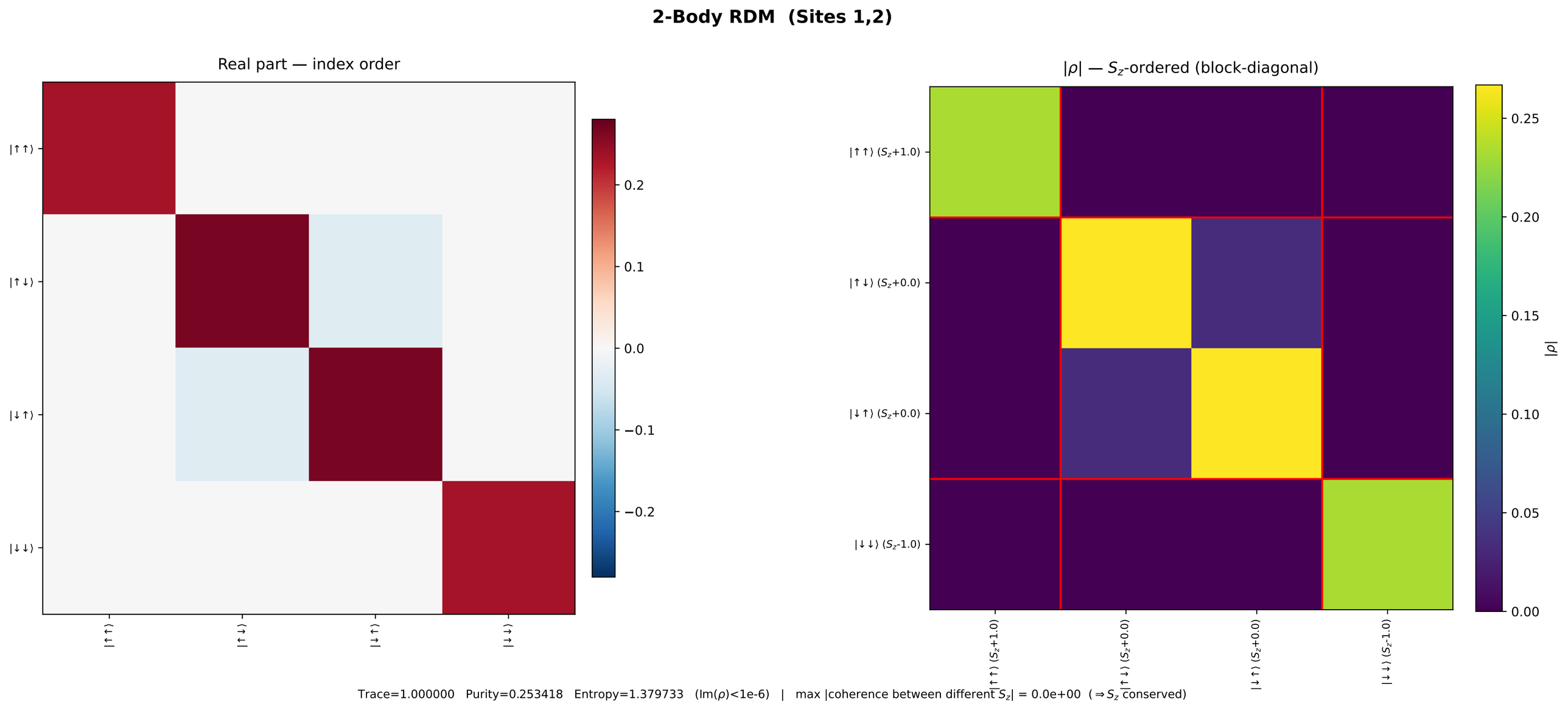

2-Body RDM (Sites 0,1)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.340081 Entropy=1.231353 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

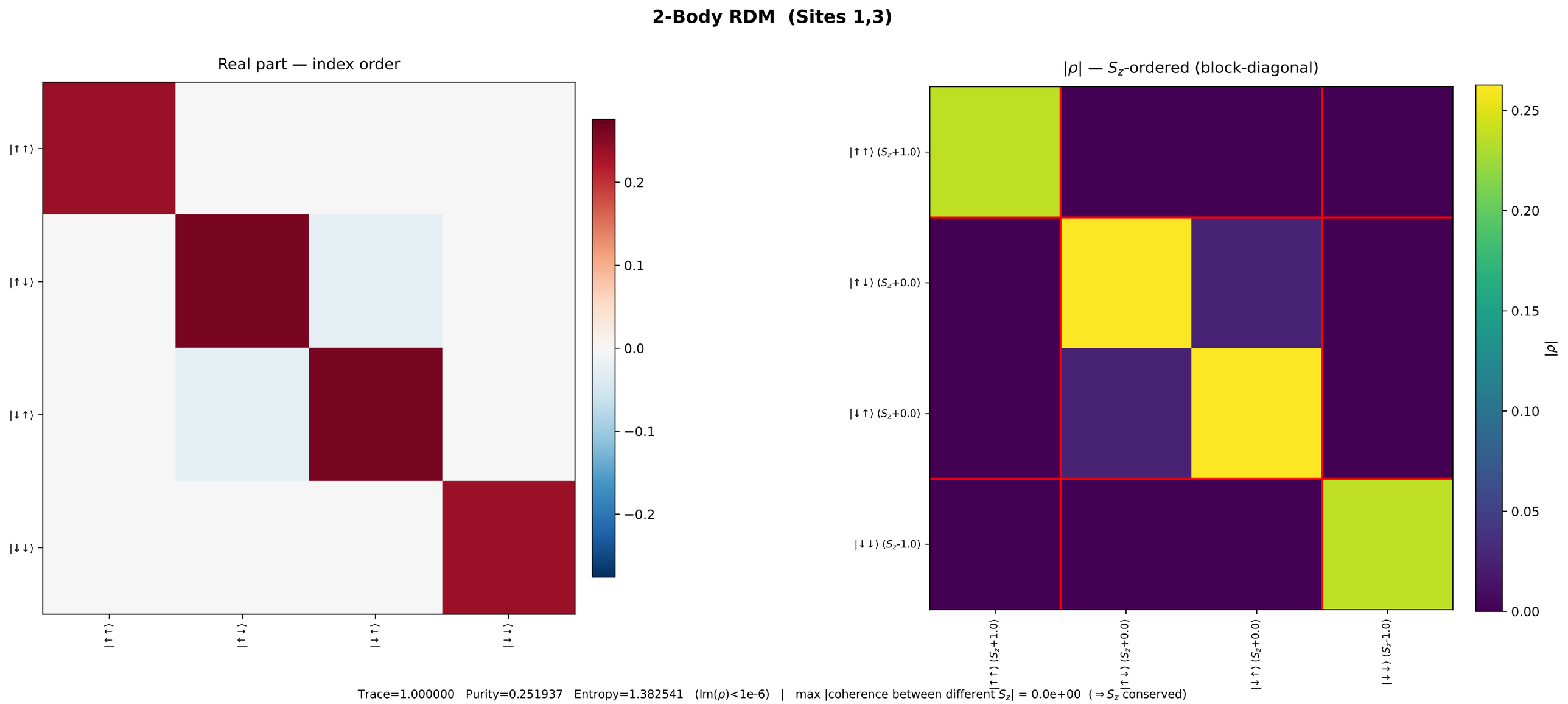

2-Body RDM (Sites 0,2)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.250120 Entropy=1.386055 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

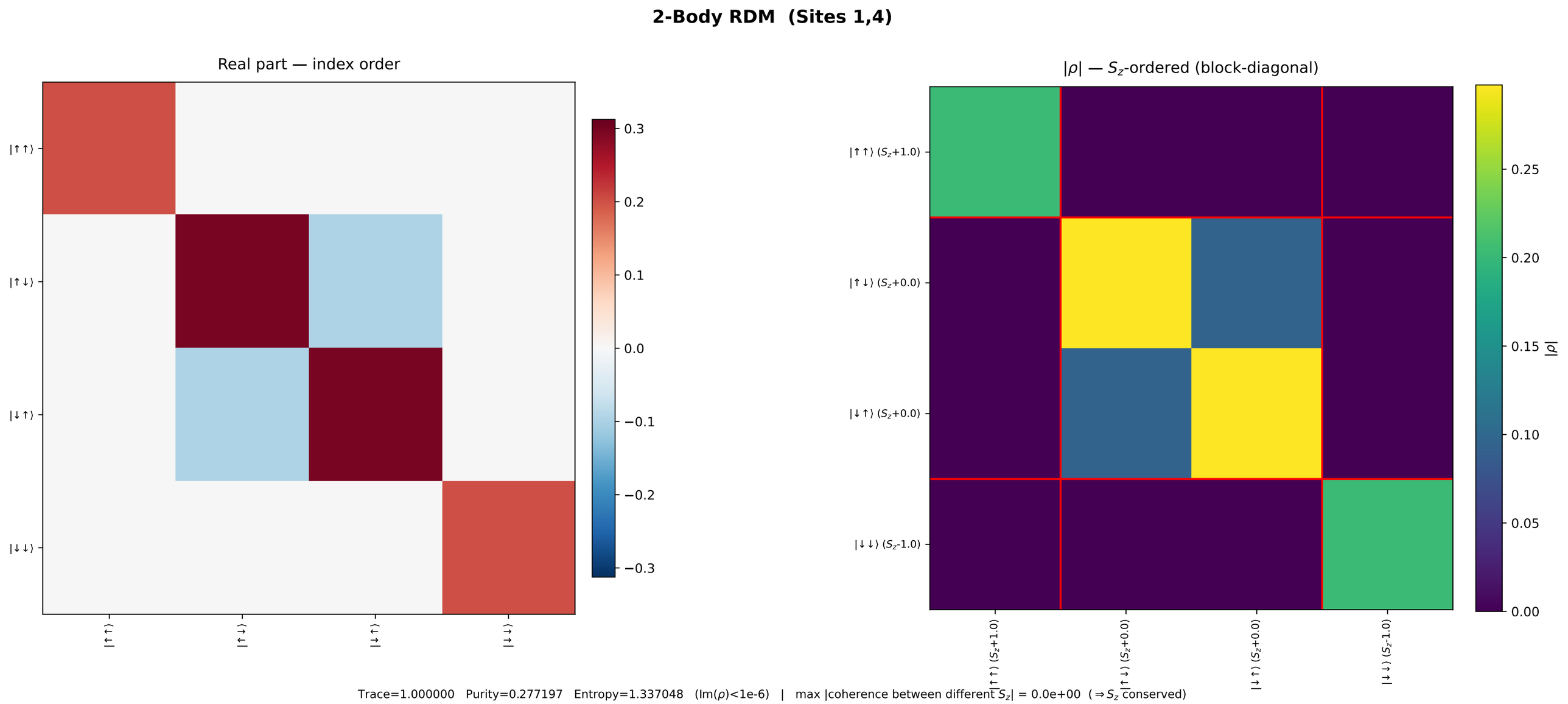

2-Body RDM (Sites 0,3)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.253738 Entropy=1.379132 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

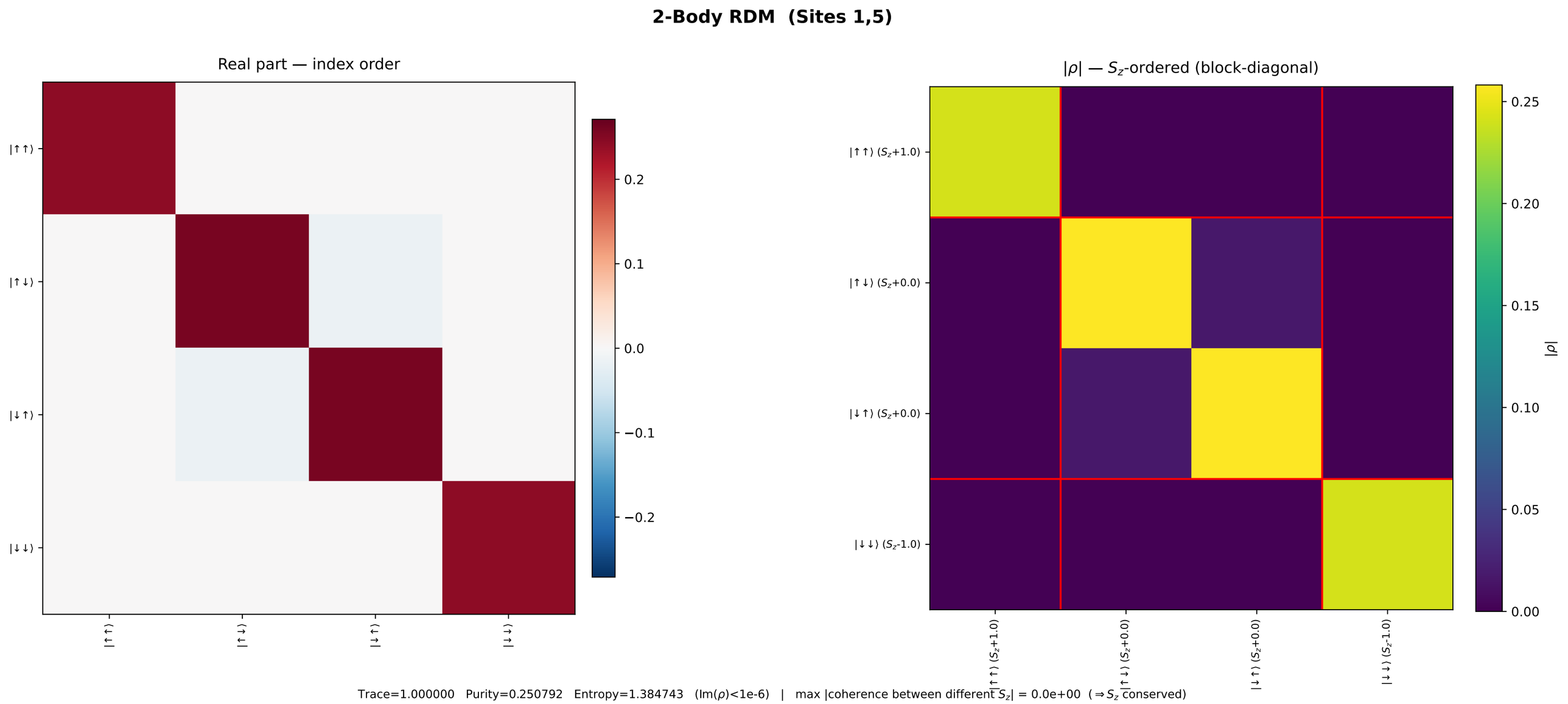

2-Body RDM (Sites 0,4)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.250055 Entropy=1.386186 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

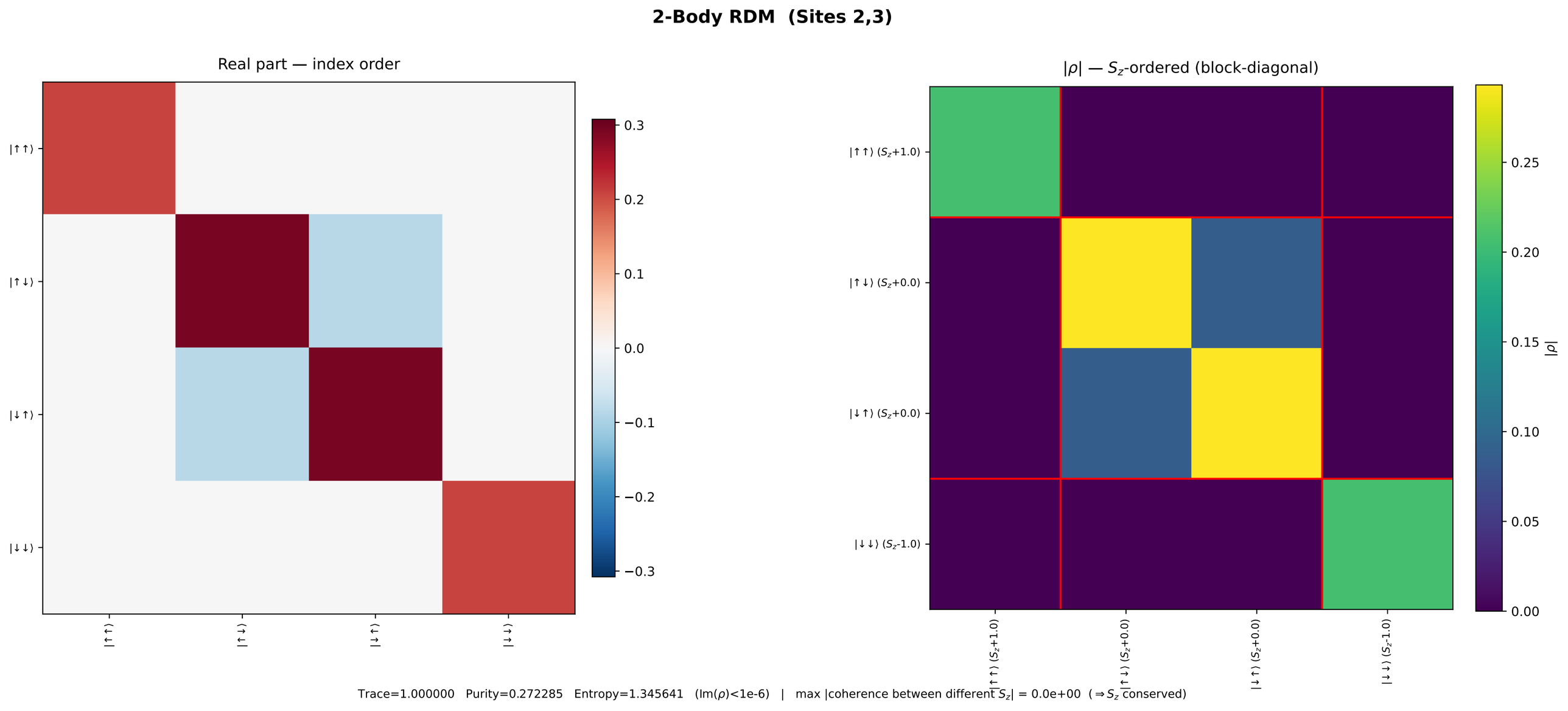

2-Body RDM (Sites 0,5)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.275580 Entropy=1.339868 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

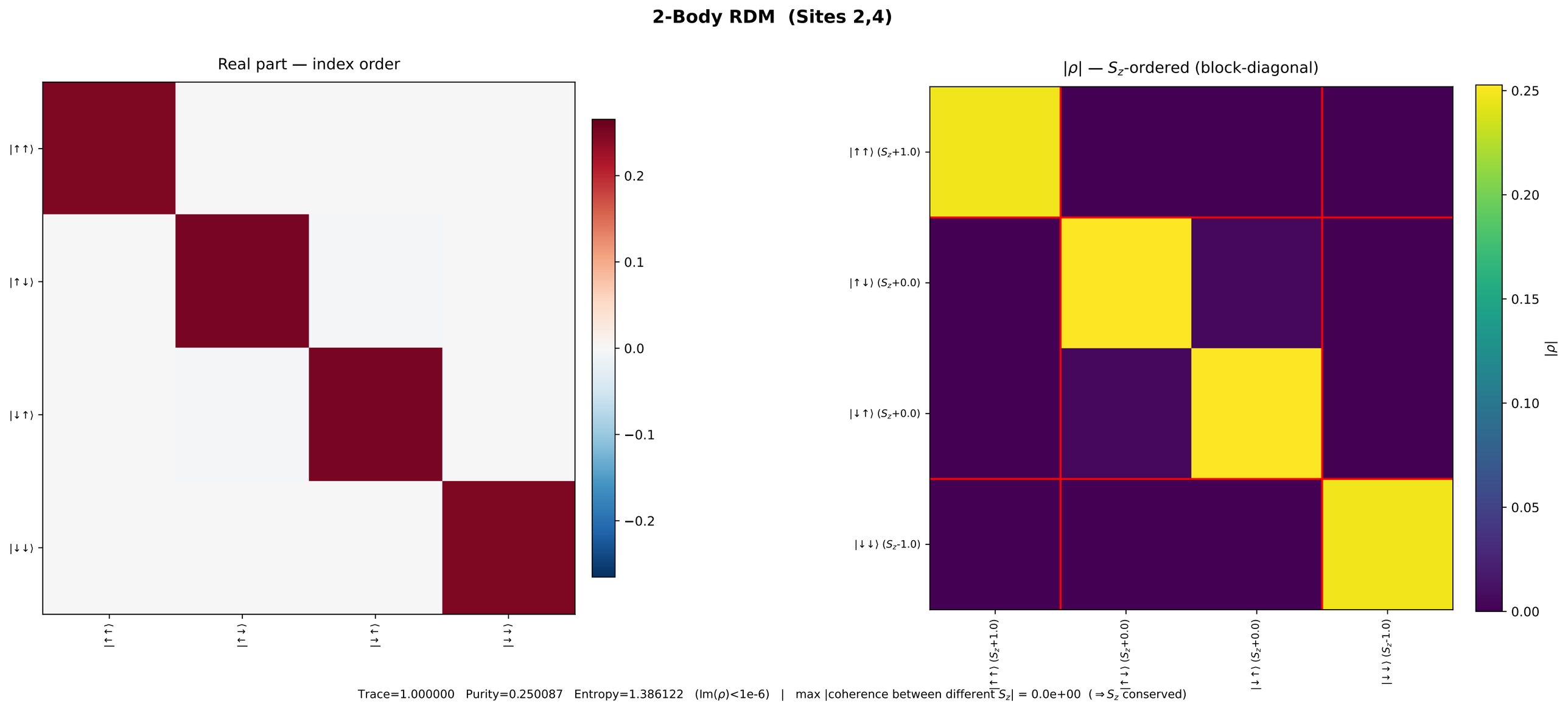

2-Body RDM (Sites 1,2)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.253418 Entropy=1.379733 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

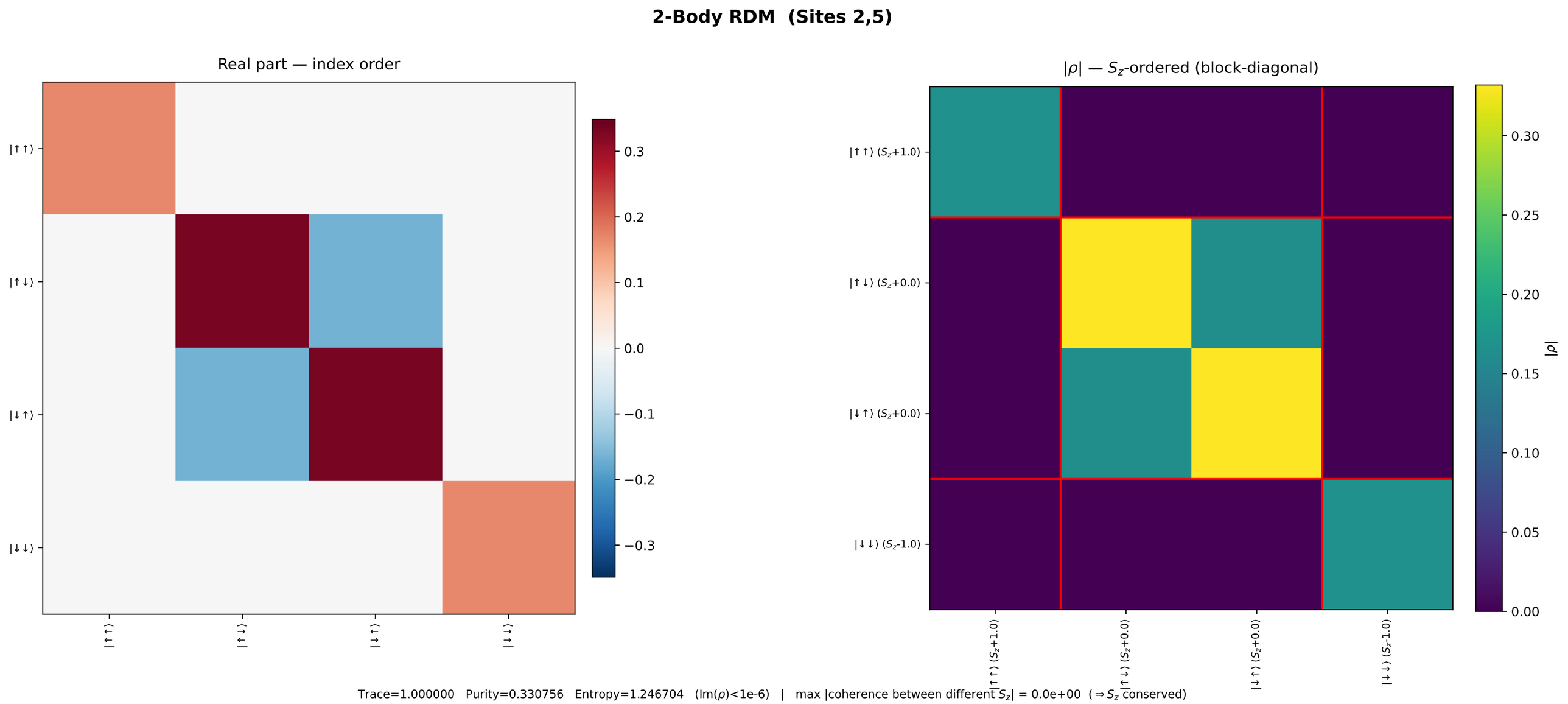

2-Body RDM (Sites 1,3)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.251937 Entropy=1.382541 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

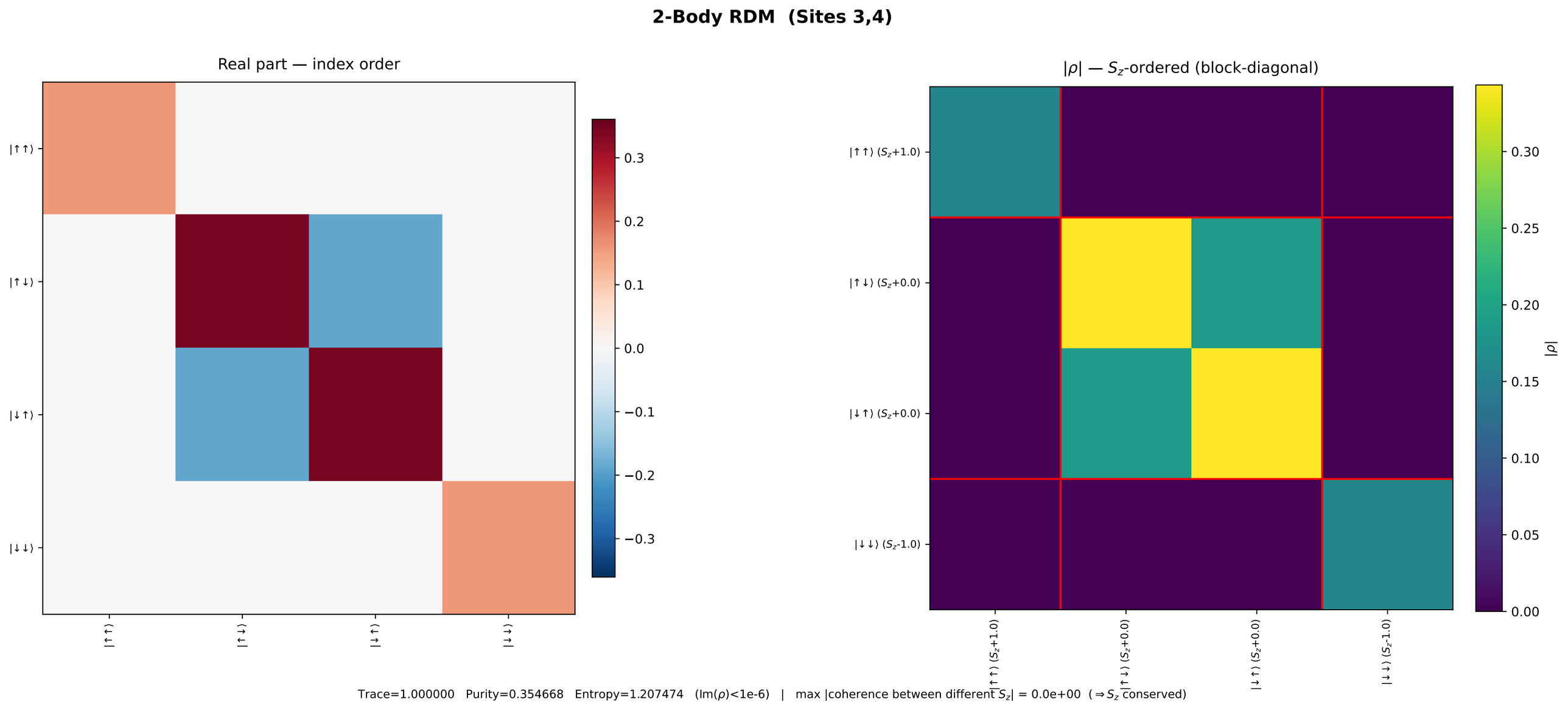

2-Body RDM (Sites 1,4)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.277197 Entropy=1.337048 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

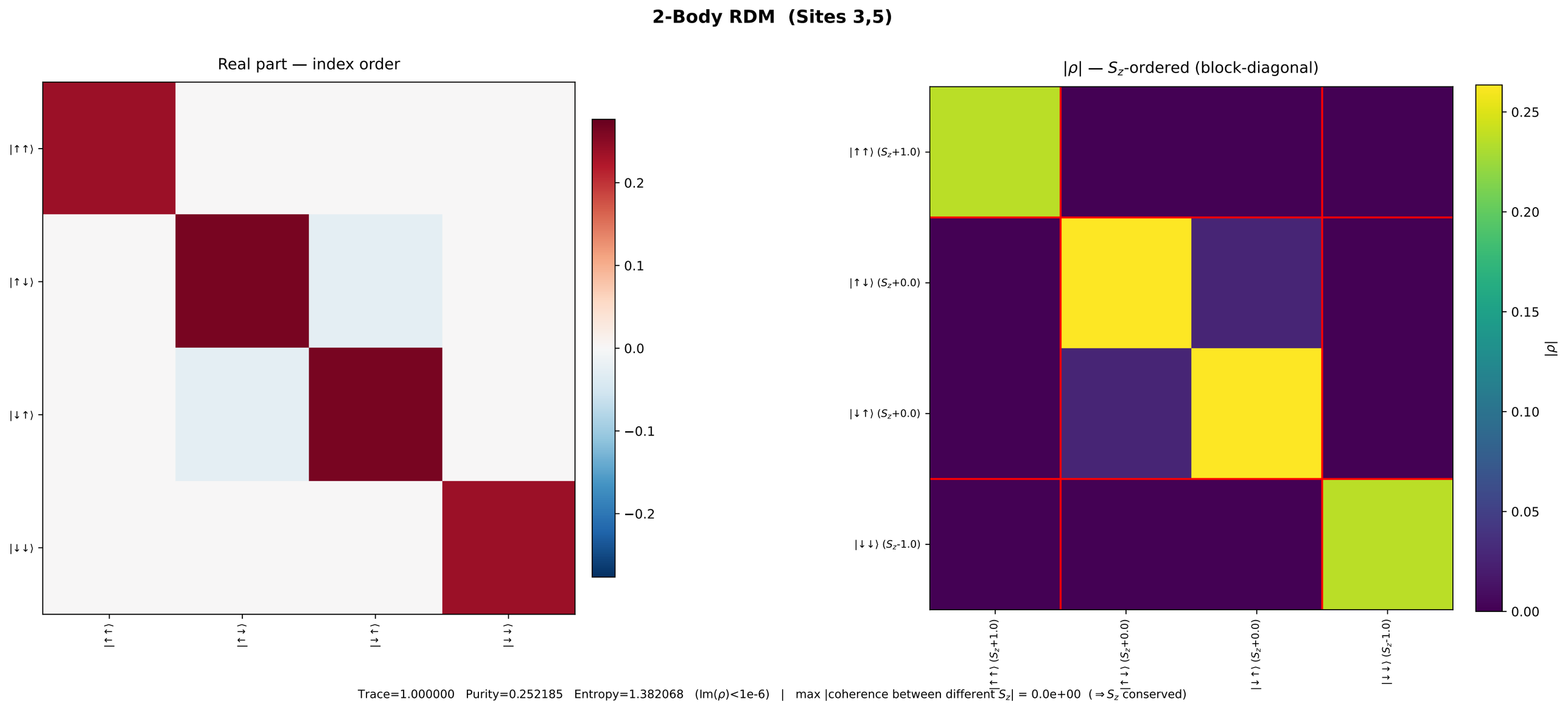

2-Body RDM (Sites 1,5)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.250792 Entropy=1.384743 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

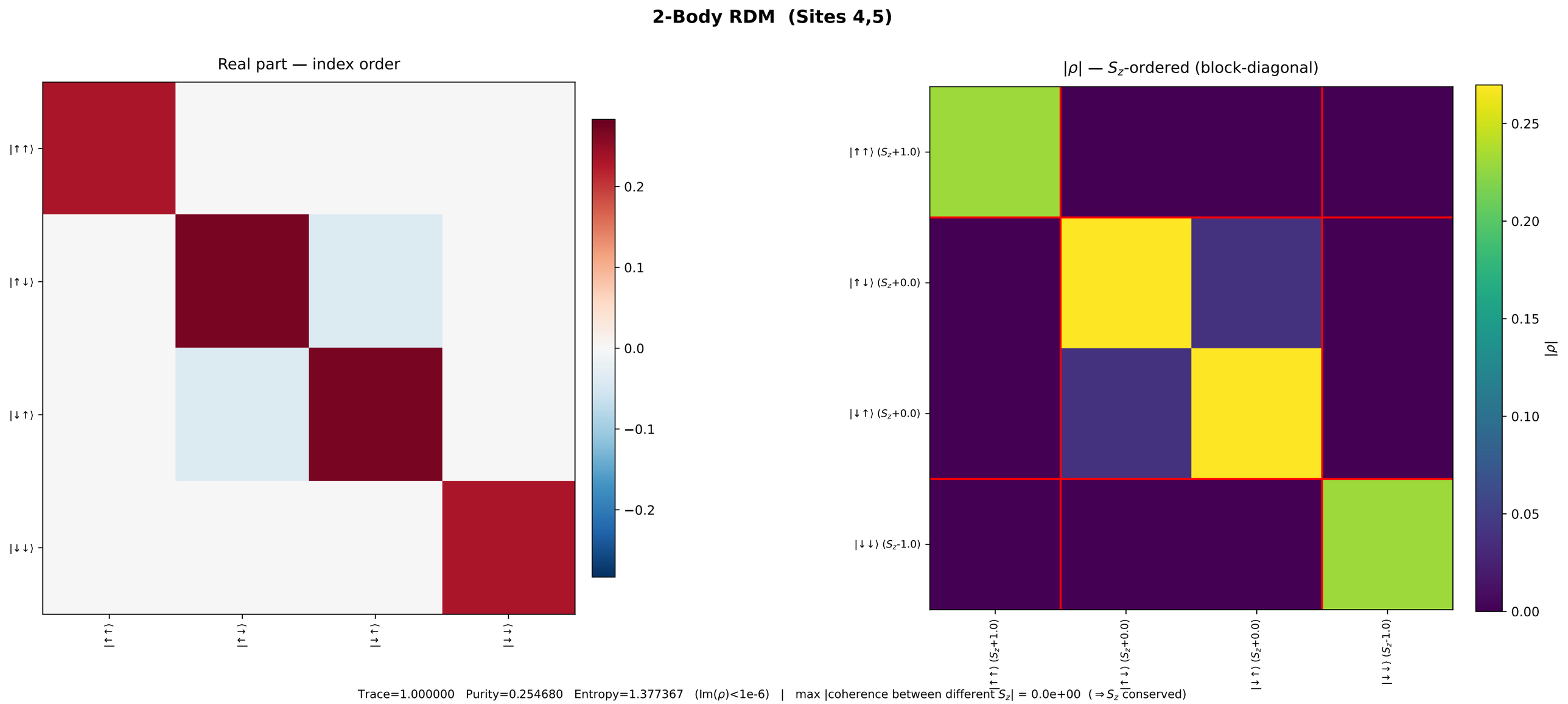

2-Body RDM (Sites 2,3)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.272285 Entropy=1.345641 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 2,4)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.250087 Entropy=1.386122 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 2,5)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.330756 Entropy=1.246704 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 3,4)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.354668 Entropy=1.207474 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 3,5)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.252185 Entropy=1.382068 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 4,5)
Real part — index order
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
|ρ| — S_z-ordered (block-diagonal)
|↑↑⟩ (S_z+1.0)
|↑↓⟩ (S_z+0.0)
|↓↑⟩ (S_z+0.0)
|↓↓⟩ (S_z-1.0)
0.25
0.20
0.15
0.10
0.05
0.00
|ρ|
Trace=1.000000 Purity=0.254680 Entropy=1.377367 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)